FIITJEE
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JEE (Advanced)-2024
PART TEST – I
PAPER –2
TEST DATE: 19-11-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D

Sol. balance the torques.

2. B

Sol. The mass on the right will oscillate because the tension in the string oscillates; however, it cannot oscillate horizontally because there is no torque exerted on it. Suppose it oscillates vertically with no net motion. Then the average y-component of the tension acting on the mass on the left is equal to mg, but the x-component of the tension is nonzero; then the average magnitude of the tension is greater than mg, a contradiction. Thus the right mass must instead oscillate and move upward.

3. A

Sol. The motion is the superposition of two motions: uniform translation of both masses with speed $\frac{m_1 v}{(m_1+m_2)}$ and circular motion about the common center of mass, where the first mass has speed $\frac{m_2 v}{(m_1+m_2)}$. The second mass comes to rest again after one period of the circular motion. The radius is $\frac{L m_2 v}{(m_1+m_2)}$, giving a period of $t = \frac{2\pi L}{v}$.

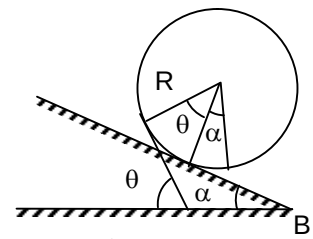
4. D

Sol. The problem can be solved using only momentum conservation, without knowing the friction between the ball and bowl. Work in the frame originally moving with the cart, where the momentum is zero. When the ball is at the bottom of the bowl, the ball and bowl have relative velocity v_b. This is only consistent with momentum conservation if the velocity of the bowl and ball are $-\frac{m}{m+M_c}v_b, \frac{M_c}{m+M_c}v_b$ respectively. Transforming back to the original frame, the final velocity of the bowl is $\frac{(m+M_c)v_0 - mv_b}{m+M_c}$

5. A, B

Sol. Let the angle formed from the mass, the center of the cylinder O, and the vertical be θ. By summing forces on the mass, we get

$$mg\sin\theta - \mu mg\cos\theta = 0 \Rightarrow \mu = \tan\theta$$



This is unsurprising, as it is the typical condition for an object to not slip. You can verify yourself that the effective angle of the incline is equal to the angle the normal force makes with the vertical, θ. Next, we sum up the torques with respect to the contact point between the ramp and the cylinder. The moment arm for the cylinder is R sin α and the moment arm for the block is R sin θ −R sin α. Therefore, we can write the torque balance equation as:

$$(M+m)g\sin\alpha = mg\sin\theta$$

Because tan θ = μ, we have $\sin\theta = \frac{\mu}{\sqrt{\mu^2+1}}$ Substituting this result into our equation of sum of torques at point P gives us

$$(M+m)g\sin\alpha = mg\frac{\mu}{\sqrt{\mu^2+1}}$$

Which gives

$$\alpha = \sin^{-1}\left(\frac{m}{(M+m)}\frac{\mu}{\sqrt{\mu^2+1}}\right) = 30°$$

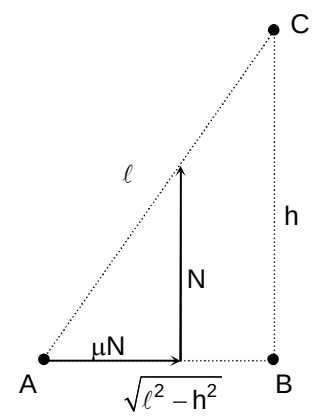
So, angle must be less than 30°

6. C, D

Sol. Consider what happens when the applied force approaches infinity. To maintain equilibrium, the friction force between the rod and the board must also increase. This friction force will also approach infinity. When dealing with large forces, we can ignore constant forces such as the weight of both the board and the rod.

As a result, since the weight of the rod is negligible we can pretend it's a mass-less rod. We also know that the forces at the ends of a massless rod will always point along the rod. For example, the force exerted on the rod by the board must point along the rod as well. The angle of this force is solely dependent on the friction coefficient μ_1. Therefore:

$$\tan\alpha < \frac{\mu_1 N}{N}, \quad \mu_1 > \frac{\sqrt{l^2-h^2}}{h}$$



7. A, B, C, D

Sol. The key insight is noting if the net vertical forces of the normal and friction forces were directed downwards then the stopper would be blocked. Let us then try to calculate the vertical components of forces that are involved. Let the normal force directed on the wedge be N.

We then know that the vertical component of the normal force is clearly either N cos α or N sin α. We can figure the component by chasing angles around, but an easier way is to imagine α → 0. In this case, the horizontal component of the normal force also goes to zero, which is the behavior of a sine function, so the horizontal component is N sin α. This in turn means that the vertical component of force involved is N cos α. We now try to find the vertical component of friction involved. The friction force directed downwards on the direction of the wedge is μN (because N is already perpendicular, you do not have to manipulate it with trigonometric functions). This means that the vertical component of friction is μN sin α. We now equate these, with an inequality where the vertical component of the normal force greater than the friction force for the wedge to pass through. N cos α > μN sin α

SECTION – B

8. 5

Sol. When the wedge has moved through a distance y, let the point of contact of cylinder have moved down the plane through x (with respect to wedge). Let F be the friction between the cylinder and the plane. Since there is no horizontal force on (wedge + cylinder system)

$$M\frac{d^2y}{dt^2}+m\left(\frac{d^2y}{dt^2}-\cos\alpha\frac{d^2x}{dt^2}\right)=0 \qquad (1)$$

Also ,

$$m\left(\frac{d^2y}{dt^2}\cos\alpha-\frac{d^2x}{dt^2}\right)=F-mg\sin\alpha$$

$$\frac{ma}{2}\frac{d^2x}{dt^2}=\frac{ma^2}{2}\frac{d^2\theta}{dt^2}=Fa$$

This gives : $2\frac{d^2y}{dt^2}\cos\alpha-3\left(\frac{d^2x}{dt^2}\right)=-2g\sin\alpha$ (2)

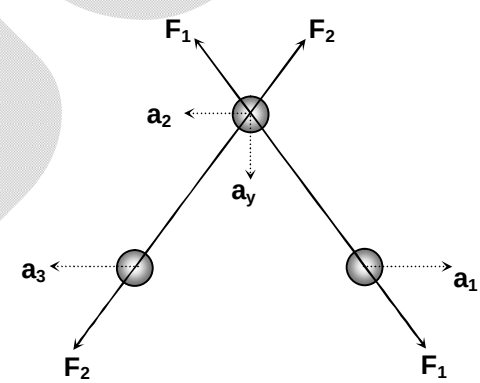
Equation (1) and (2) implies acceleration of the centre of the cylinder down the face, and relative to the wedge, is

$$\frac{2g\sin\alpha(M+m)}{3M+m+2m\sin^2\alpha}=5\frac{m}{s^2}$$

9. 1

Sol. There is no external force in the horizontal direction, therefore:

$$ma_2+ma_3=4ma_1$$



The component of acceleration of the top mass along the left rod must be the same as the component of acceleration of the left mass along the left rod. This means:

$$a_2\cos45^0+a_y\sin45^0=a_3\cos45^0$$

$$\Rightarrow a_2+a_y=a_3$$

The same is true along the right rod, giving

$$a_y-a_2=a_1$$

Solving these three equations, we get:

$$a_y=2a_1$$

$$a_2=a_1$$

$$a_3=3a_1$$

Using Newton's Second Law on the rightmost mass, we have

$$F_1\cos45^0=4ma_1$$

For the left mass

$$F_2\cos45^0=ma_3=3ma_1$$

For the top mass in the vertical direction. We get:

$$mg-(F_1\cos45^0+F_2\cos45^0)=ma_y=2ma_1$$

Substituting in F_1 and F_2 from above gives

$$mg-7ma_1=2ma_1$$

$$\Rightarrow a_1=\frac{g}{9}=1$$

10. 2

Sol. Let's denote the speed of the small body at the top point of the semicircle v_1 .Since leaving the half circle, the body makes a horizontal throw starting from height 2 R , during this time it travels a distance L in horizontal direction, it can be written that

$$2R = \frac{gt^2}{2} \text{ and } L = v_1 t$$

from where $v_1^2 = \frac{L_1 g}{4R}$

Comment. This is the speed at the moment before leaving the forced orbit

$$a = \frac{v_1^2}{R} = \left(\frac{L}{2R}\right)^2 g = 4g$$

corresponds to centripetal acceleration. Since a > g , the body neither then nor before leaves the semicircular path, it slides along it.

applying the work energy theorem:

$$\frac{1}{2}mv_1^2 - \frac{1}{2}mv_0^2 = -2\mu mgL - 2Rmg \Rightarrow v_0 = \sqrt{\left(\frac{L^2}{4R} + 4\mu L + 4R\right)g} \approx 8.4\frac{m}{s}$$

11. 3

Sol. COM will be at $\frac{3\pi R}{16}$

12. 4

Sol. COM will be at $\frac{\pi R}{4}$ from centre

13. 12

Sol. $v_o \cos 30^0 = v \cos 45^0$

$$\Rightarrow v = v_o\sqrt{\frac{3}{2}}$$

Now using conservation of energy

$$\frac{1}{2}m\left(v^2 - v_0^2\right) = mgh$$

SECTION – C

14. 600.00

15. 1130.00

Sol. (Q.14-15):

Cruising speed of train is 72 km/h = 20 m/s .Train moving with an acceleration of 1 m/s^2 reaches its maximum speed in time $t_1 = v_0/a$ =20 sec and needs the same amount of time to decelerate. During this time, the train travels 400 m .

Total number of stops = 7

Total number of journeys = 8

Total distance covered by the train while accelerating or decelerating = 400 x 8 = 3200m = 3.2 km

Distance travelled at cruising speed = 15.2 km – 3.2 km = 12 km

(During the calculation, we did not assume that the distance between the stations is the same, which is not true anyway.)

Total time covered by the train while accelerating or decelerating = 40x8 = 320 sec

Time for which the train travelled at cruising speed = $\frac{12000}{20} = 600\,\text{sec}$

Total time of journey between Delhi and Gurugram = 600+320+(30x7)= 1130 sec

16. 13.00

Sol. When the block is at rest relative to the box, they have the same (final) velocity v relative to the lab frame. Since there are no external forces, the total momentum is conserved and we have, $2mv = mv_0 \Rightarrow v = \frac{v_0}{2}$

Since the box and the block have the same mass and the collision is elastic, the relative velocity of the block relative to the box change sign after every collision and the block decelerates between collisions due to the friction.

Hence, the total distance travelled by the block before it stops (relative to the box) is,

$$\mu mgs = \frac{1}{2}mv_0^2 - 2x\frac{1}{2}mv^2 \Rightarrow s = \frac{v_0^2}{4\mu g} = 12.76m$$

The block collides with the box at distances 0.5 m, 1.5 m, ..., 12.5 m. Hence the number of collisions is 13.

17. 12.64

Sol. Since there are no external forces acting on the system, the center of mass (CM) always moves with the constant velocity v. Hence the distance travelled by the CM is $x_{cm} = vt = \frac{v_0 t}{2}$. where t is the time for the block to decelerate from v_0 to v.

Consider the frictional force acting on the block, we have

$$-\mu mgt = mv - mv_0 = -\frac{1}{2}mv_0 \Rightarrow t = \frac{v_0}{2\mu g} = 5.1s$$

$$x_{cm} = \frac{v_0 t}{2} = 12.76m$$

Initially, B is at the midpoint of A. At distances 0.5 m, 2.5 m, ..., 12.5 m, the block collides with the right wall of the box. Hence when the block stops, it lies at 0.26 m from the right wall. So finally it will stop at 0.24 m to the right of the midpoint. Hence the CM is shifted to 0.12 m to the right of the midpoint when the block B is at rest relative to the box. The distance of the box travelled is

$$x_A = x_{cm} = 0.12 = 12.64m$$

Chemistry

PART – II

SECTION – A

18. D
Sol. (I) Na does not react with N_2 to form Na_3N.
(II) LiH is more stable than NaH.

19. C
Sol. In Borax two Boron atoms are sp^3 hybridised and two Boron atoms are sp^2 hybridised.

20. D
Sol. Hybridisation of Br in $\left[BrF_4\right]^-$ is sp^3d^2.

21. B
Sol. (I) $BaSO_4$ is less soluble in water than $CaSO_4$.
(II) $2Mg\left(NO_3\right)_2 \xrightarrow{\Delta} 2MgO + 4NO_2 + O_2$

22. C, D
Sol. $K_P = \left(P_{NH_3}\right)^2\left(P_{CO_2}\right)$

23. C, D
Sol. (I) Solubility of AgCN in NH_3 solution will be higher due to complex formation.
(II) AgCN is more soluble in acidic buffer solution.

24. B, D
Sol. $\log k = \log A - \frac{E_a}{2.303RT}$
log A = 4,
$A = 10^4$
$\frac{E_a}{2.303R} = 3000$
$E_a = 3000 \times 2.303 \times 2$
= 13818 cal
= 13.818 Kcal

SECTION – B

25. 3
Sol. SF_3Cl, XeO_2F_2, SF_4

26. 4
Sol. N_2^-, H_2^-, N_2^+, O_2^+ are paramagnetic.

27. 4
Sol. $Zn(OH)_2$, $Al(OH)_3$, $Pb(OH)_2$ and $Sn(OH)_2$ are amphoteric, hence they are soluble in excess of NaOH solution.

28. 50

Sol. Milli moles of $H_2SO_4 = \frac{4.9}{98} = 0.05$

Milli moles of $HCl = \frac{3.65}{36.5} = 0.1$

Milli moles of $HNO_3 = \frac{3.15}{63} = 0.05$

$\therefore \left[H^+\right] = \frac{(2\times 0.05)+0.1+0.05}{V(ml)}$

$5\times 10^{-5} = \frac{0.25}{V(ml)}$

$V(ml) = \frac{0.25}{5\times 10^{-5}} = 5000$ ml

$\therefore \frac{V}{100} = \frac{5000}{100} = 50$

29. 56

Sol.

	$A(g) \longrightarrow$	$2B(g)$	$+C(g)$
$t = 0$	P^o	0	0
$t = 5$ min	$P^o - x$	$2x$	x
$t = \infty$	0	$2P^o$	P^o

$\therefore$ when $t = \infty$, $3P^o = 240$ mm

$\therefore$ $P^o = 80$ mm

At t = 5, $P^o - x + 2x + x = 128$

$P^o + 2x = 128$

$80 + 2x = 128$

x = 24

Partial pressure of A(g) at the end of 5 min $y = P^o - x = 80 - 24 = 56$ mm

$\therefore y = 56$

30. 6

Sol. $9SO_2 + 3K_2Cr_2O_7 + 3H_2SO_4 \longrightarrow 3K_2SO_4 + 3Cr_2(SO_4)_3 + 3H_2O$

$x = 3, \quad y = 3$

$x + y = 3 + 3 = 6$

SECTION – C

31. 41.25

Sol. $\frac{hC}{\lambda} = (2.3 + 0.70)\times 1.6\times 10^{-19}$ J

$\lambda = \frac{6.6\times 10^{-34}\times 3\times 10^{8}}{3\times 1.6\times 10^{-19}}$

$\lambda = 41.25\times 10^{-8}$ m

$\therefore y = 41.25$

32. 19.95

Sol. $\frac{hC}{\lambda} = h\nu_0 + KE_{max}$

$KE_{max} = \frac{6.6\times10^{-34}\times3\times10^{8}}{500\times10^{-10}\times1.6\times10^{-19}} - 4.8$

$= 24.75 - 4.8$

= 19.95 eV

33. 47.52

Sol. $Ca_2B_6O_{11} + 11H_2O + 4SO_2 \longrightarrow \underset{(P)}{2Ca(HSO_3)_2} + \underset{(Q)}{6H_3BO_3}$

$2H_3BO_3 \xrightarrow{\text{Strong heating}} \underset{(R)}{B_2O_3} + \underset{(S)}{3H_2O}$

$B_2O_3 + P_2O_5 \longrightarrow \underset{(T)}{2BPO_4}$

Compound P is $Ca(HSO_3)_2$

% mass of oxygen $= \frac{96}{202}\times100 = 47.52$

34. 6.00

Sol. Compound 'T' is BPO_4.

Mathematics

PART – III

SECTION – A

35. C

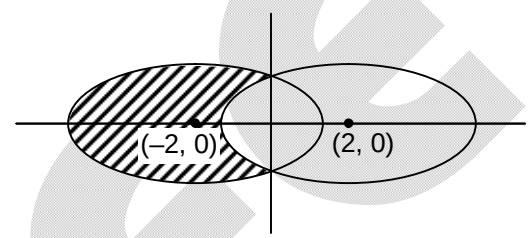
Sol. $h'(0)=\left(\lim_{h\to 0}\frac{g(0+h)-g(0)}{h}\right)\left(e^{g(0)}\cdot f^2(0)g(0)+e^{g(0)}f^2(0)\right)+2e^{g(0)}f(0)f'(0)g(0)$

For h′(0) to be exist either f(0) = 0 or g(0) = –1

36. C

Sol. Point P moves inside the ellipse $\frac{(x+2)^2}{9}+\frac{y^2}{5}=1$

and outside the ellipse $\frac{(x-2)^2}{9}+\frac{y^2}{5}=1$



Required area is shown in the figure

$= 3\sqrt{5}\pi-4\int_0^1\sqrt{5}\sqrt{\left(1-\frac{(x+2)^2}{9}\right)}\,dx$

$= 3\sqrt{5}\pi-\frac{4}{3}\sqrt{5}\int_0^1\sqrt{9-(x+2)^2}\,dx = 3\sqrt{5}\pi+\frac{20}{3}-3\sqrt{5}\pi+6\sqrt{8}\sin^{-1}\frac{2}{3} = \frac{20}{3}+6\sqrt{5}\sin^{-1}\left(\frac{2}{3}\right)$

37. C

Sol. $\sum_{r=1}^{n}\operatorname{cosec}\frac{x}{2^{r-1}}=\cot\frac{x}{2^n}-\cot x$

Hence, $f_n(x)=\tan^{-1}(-\cot x)$

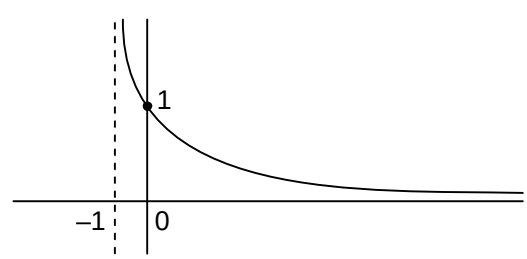
$f_5\left(\frac{\pi}{4}\right)=\tan^{-1}(-1)=-\frac{\pi}{4}$

$f_5\left(\frac{3\pi}{4}\right)=\tan^{-1}(1)=\frac{\pi}{4}$

38. D

Sol. $\frac{\log_e(1+x)}{x}$

is a decreasing function and its graph is as follows



39. A, B, C

Sol. $\lim_{x\to 2^+}f(x)=1,\ \lim_{x\to 2^-}f(x)=-1,\ \lim_{x\to 1^+}f(x)=-1$

40. A, B, C

Sol. $f(x) + f(x^2) + f(x^3) = f(x + x^2 + x^3)$ differentiate w.r.t x, we get

$f'(x) + 2xf'(x^2) + 3x^2f'(x^3) = (1 + 2x + 3x^2)f'(x + x^2 + x^3)$

Put $x = 1 \Rightarrow 6f'(1) = 6f'(3) \Rightarrow f'(1) = f'(3)$

Put $x = -1 \Rightarrow 4f'(-1) - 2f'(1) = 2f'(-1) \Rightarrow f'(-1) = f'(1)$

Put x = 1 in given relation 3f(1) = f(3)
Put x = –1 in given equation 2f(–1) + f(1) = f(–1) $\Rightarrow$ f(–1) + f(1) = 0
$\therefore$ 3f(–1) = –f(3)

41. B, C

Sol. f(x) = sin x $\rightarrow$ periodic
g(x) = x + sin x $\rightarrow$ non-linear and non-periodic
f(g(x)) = sin(x + sin x) $\rightarrow$ periodic with period 2π
Hence, option (A) is incorrect and option (B) is correct

$f(x) = \sin x^{\frac{1}{3}}$ non-periodic

$g(x) = x^3$ non-periodic

f(g(x)) = sin x periodic with period 2π
Hence, option (C) is correct but option (D) is incorrect

SECTION – B

42. 0

Sol. Let $\angle AOQ = \theta$ and radius of circle = r

Area of sector AOQ = $\frac{1}{2}r^2\theta$

Area of $\Delta AOP = \frac{1}{2}r \cdot r\tan\theta = \frac{r^2}{2}\tan\theta \Rightarrow$ Area of curvilinear triangle APQ = $\frac{r^2}{2}(\tan\theta - \theta)$

Also, $A \rightarrow Q \Rightarrow \theta \rightarrow 0$. Now, $\lim\limits_{A\to Q}\frac{S_2}{S_1} = \lim\limits_{\theta\to 0}\frac{\tan\theta - \theta}{\theta} = \lim\limits_{\theta\to 0}\left(\frac{\tan\theta}{\theta} - 1\right) = 0$

43. 2

Sol. Area of triangle will be $A = \frac{r}{16}(16 - r^2)^{\frac{3}{2}}$ for maximum area $\frac{dA}{dr} = 0 \Rightarrow r = 2$

44. 1

Sol. By using AM $\geq$ GM, we get maximum value of f(x) = 1

45. 7

Sol. $f(x) = x\sqrt{1 + f(x+1)}$ by solving, we get f(x) = x(x + 2)

46. 0

Sol. $f(x) = \ln\left|\left(e^{1/x} - e^{-1/x}\right)\right|$

$\lim\limits_{x\to\infty} f(x) = 0$

47. 0

Sol. $\frac{dy}{dx} = \frac{1 - \ln x}{x^2 + \ln^2 x} \Rightarrow y = \int \frac{1 - \ln x}{x^2 + \ln^2 x}dx$

$\Rightarrow y = \int \frac{1}{1 + \left(\frac{\ln x}{x}\right)^2} \cdot \left(\frac{1 - \ln x}{x^2}\right)dx \Rightarrow y = \tan^{-1}\left(\frac{\ln x}{x}\right) + c$

Given $y(1) = \frac{\pi}{2} \therefore \frac{\pi}{2} = c$

Now, $y\left(\frac{1}{e}\right) = \tan^{-1}(-e) + \frac{\pi}{2} = -\tan^{-1}e + \frac{\pi}{2}$

$\Rightarrow \left[y\left(\frac{1}{e}\right)\right] = 0$

SECTION – C

48. 4.00

49. 1.14

Sol. (Q.48.-49.):

If $\angle AOB = \theta$, then for S_{max} $\theta = \frac{\pi}{2}$ and $S_{max} = 4$

Also, in this case area which is common to ΔECD and inner circle will be = $\frac{4\pi}{4} - 2 = 1.14$

50. 12.56

51. 0.00

Sol. (Q.50.-51.):

f(g(x)) = sin(x – sin x) period = 2π

g(f(x)) = sin x – sin(sin x) period = 2π

Hence, $T_1 + T_2 = 12.56$

$$\int_0^{2\pi} \left(\sin(x - \sin x) + \sin x - \sin(\sin x)\right) dx$$